# Causal normalizing flows: from theory to practice

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## Why We Bridge Causal ML and Generative Modeling

- We study what-if and counterfactual reasoning that drives decisions in healthcare, policy, and fairness.
- We leverage normalizing flows for exact likelihoods and tractable inverses on complex data.
- We aim to make causal inference practical with modern deep generative modeling.



## The Problem: How We Learn Causal Models and Interventions from Observations

- We seek identifiable causal models from observational data under minimal assumptions.
- We avoid brittle per-node models and heuristic graph surgery for interventions.
- We handle mixed data and partial graphs while remaining scalable and accurate.



## Our Core Idea and Contributions (High-Level)

- We unify SCMs and autoregressive NFs via triangular monotone increasing maps and prove identifiability.
- We design causally consistent NF architectures and show a single-layer abductive ANF works best.
- We implement a general do-operator by manipulating exogenous variables for interventions and counterfactuals.
- We extend to mixed discrete–continuous data and partial graphs (SCC-based), and validate extensively.

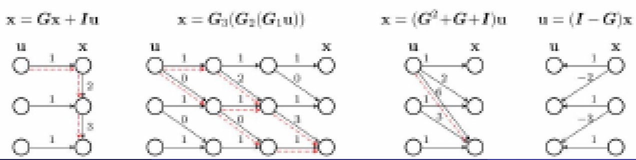
Takeaway: We learn an identifiable single-flow model aligned with the causal graph that supports sound, efficient interventions and counterfactuals.

## From SCMs to Triangular Flows: Why Structure Matters

- We exploit SCM-induced causal orderings to obtain triangular maps from exogenous $u$ to observed $x$.
- We align triangular, monotone structure with autoregressive NFs for exact likelihoods and tractable inverses.
- We control Jacobian sparsity via the graph, linking flow dependencies to causal edges.

### Key pieces

- $u$ are independent exogenous variables.
- $f : u \to x$ is a triangular flow aligned with the causal order.
- Masks $M(A)$ enforce parent-only dependencies.
- Jacobian sparsity links the flow to the causal graph.






## How We Enforce Causal Graphs in Flows

- We use a topological order $\pi$ and adjacency $A$ to define valid parent dependencies.
- We enforce masks in the Jacobian to avoid spurious paths inconsistent with the graph.
- We ensure causal consistency at the optimum by respecting these structural constraints.

$$x_1 \to x_2 \to x_3$$

$$\pi = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Figure: Causal graph with an ordering $\pi$ and adjacency $A$ that guide masking in the flow.



## Identifiability: How We Isolate Exogenous Variables with Flows

- If an ANF matches the observational distribution, it recovers exogenous variables up to component-wise invertible transforms.
- At optimum, we obtain causal consistency: Jacobian sparsity matches the true causal graph.
- This connects nonlinear ICA insights to causal flow learning.

### Key components and roles

- $T_\theta(x)$ (abductive map): maps $x \to u$ with triangular Jacobian masked by $(I - A)$.
- $f$ (generative map): maps $u \to x$; triangular structure enforces parent-only dependencies.
- $P_u$ (base): fully factorized exogenous distribution; independence drives identifiability.
- Causal consistency: at optimum, the Jacobian sparsity of $T_\theta$ matches the true graph $(I - A)$.

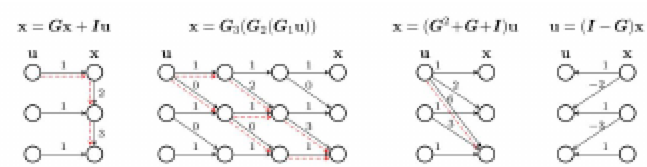


## Design Choices: Generative (u→x) vs Abductive (x→u)

- We find generative $u \to x$ requires multiple masked layers; $L \geq$ graph diameter to avoid shortcuts.
- We use abductive $x \to u$ with $L = 1$ and graph masking, which is causally consistent and easy to optimize.
- With only ordering known, we learn sparsity via Jacobian regularization on forbidden entries.

### What the masks do

- Graph mask $M(A)$: allows edges only from parents; prevents shortcuts.
- Ordering-only case: we regularize forbidden Jacobian entries to learn sparsity.
- Abductive $x \to u$ with $L = 1 + M(A)$: causally consistent by construction.




(a) Recursive. (b) Unrolled. (c) Compacted. (d) Inverted.

## How We Implement the Do-Operator in Flows

- 1) Abduction: map $x$ to $u = T_\theta(x)$.
- 2) Intervention: set the target $x_i = \alpha$ and adjust the corresponding exogenous $u_i$ so that $f_i(u) = \alpha$ (Dirac constraint on $u_i$).
- 3) Prediction: map the modified $u$ back through $f$ ($u \to x$) to sample from $p(x \mid \mathrm{do}(x_i = \alpha))$.
- Note: We operate on $P_u$ (not on structural equations) and avoid heuristic graph surgery.



## Validating the Do-Operator: Simpson's Scenario

- We observe correct path breaking after intervention, as predicted by the graph.
- We see ancestor–descendant dependencies adjust consistently with theory.
- We confirm principled interventional behavior of causal NFs.

Setup: SIMPSON NLIN; $\mathrm{do}(x_3 = -1.09)$.

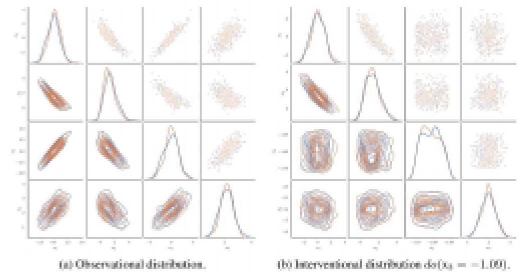



Figure: Observed vs. interventional samples in a Simpson's paradox setup; intervention severs non-causal dependencies.



## How We Handle Mixed Discrete–Continuous Data

- We use principled dequantization (add noise $\varepsilon \in [0, 1]$) to embed discrete variables into flows.
- We rely on recoverability guarantees to preserve the original discrete distribution.
- We enable causal NFs on real-world datasets with categorical features.



## How We Use SCCs When the Graph Is Partially Known

- We group nodes with unknown directions into strongly connected components (SCCs).
- We model intra-SCC with a general NF or ANF and preserve the inter-block DAG structure.
- We obtain block-wise identifiability and causal consistency at the SCC level.

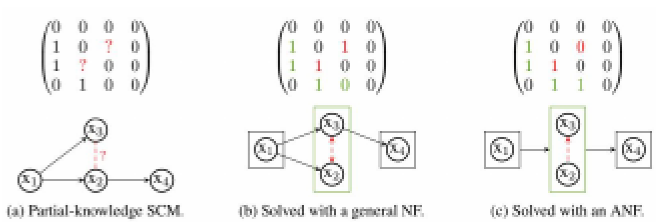



Figure: SCC reduction for partial graphs: intra-SCC modeling with preserved inter-SCC DAG.

## Partial Graphs: Ambiguity and Why We Use SCCs

- We note uncertain directions (e.g., $x^2 \leftrightarrow x^3$) can imply different causal stories, including confounding.
- We use SCC grouping to avoid premature commitment to a single direction.
- We support robust learning while respecting known structure.

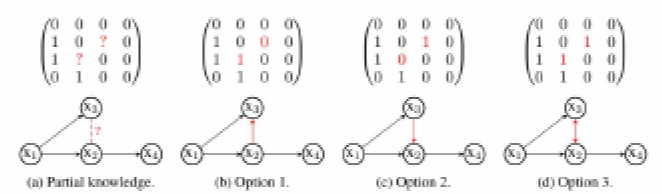



Figure: Ambiguity in partial graphs motivates SCC blocks when directions are uncertain.



## Experimental Setup: Datasets, Graphs, and Metrics

- Datasets: synthetic SCMs (chains, fork, collider, triangle, Simpson, LARGEBD) and German Credit.
- Metrics: exact likelihood (KL), ATE RMSE, counterfactual RMSE; runtime for training/eval/sampling.
- Baselines: CAREFL (Author–Year) and VACA (Author–Year).

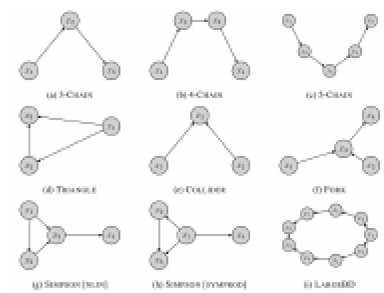


Figure: Causal graphs for the SCM benchmarks used in our experiments.



## Design Choices, Properties, and Complexity (Table 1)

- We use abductive x→u with $L = 1$ and known $A$ for causal consistency and easy optimization.
- Generative u→x respects $A$ but needs depth; sampling is faster, evaluation is slower.
- Abductive trains efficiently (evaluation) and remains accurate for causal queries.

Table: Summary of design choices, induced properties, and time complexity for density evaluation and sampling.

| Design Choices | Assumption | Model Properties | | Causal Consistency | Time Complexity | |
|---|---|---|---|---|---|---|
| | | u→x | x→u | | Sampling | Evaluation |
| Generative | Ordering | × | × | | O(L) | O(dL) |
| u→x Generative | Graph $A$ | ✓ | × | | O(L) | O(dL) |
| Abductive (L > 1) | Ordering | × | × | | O(dL) | O(L) |
| Abductive (L > 1) x→u | Graph $A$ | × | × | | O(dL) | O(L) |
| Abductive (L = 1) | Graph $A$ | ✓ | ✓ | ✓ | O(dL) | O(L) |



## Ablation: Depth, Masking, and Regularization

- We observe generative $u \to x$ needs $L \geq$ diameter; abductive $x \to u$ works well with $L = 1$.
- We find graph masks significantly improve interventional and counterfactual accuracy.
- We use Jacobian regularization when only ordering is known; it is unnecessary with full masks.

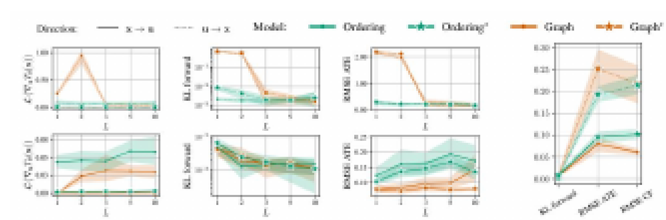


Figure: Ablation on a 4-node chain: effect of flow depth for generative vs. abductive models.

## Benchmark Comparisons: Accuracy and Speed

- We match or improve KL vs. CAREFL (Author–Year) and strongly improve ATE/CFRMSE; VACA (Author–Year) is slower and less accurate.
- We observe training/evaluation times favor causal NFs across datasets.

Table: Comparison, on different SCMs, of the proposed Causal NF, VACA, and CAREFL with the do-operator.

| Dataset | Model | KL | ATERMSE | CFRMSE | Training | Evaluation | Sampling |
|---|---|---|---|---|---|---|---|
| 3-CHAIN LIN | Causal NF | 0.000.00 | 0.050.01 | 0.040.01 | 0.410.06 | 0.480.10 | 0.760.06 |
| | CAREFL | 0.000.00 | 0.200.13 | 0.200.09 | 0.680.24 | 0.970.33 | 1.940.77 |
| | VACA | 4.441.03 | 5.760.07 | 4.980.10 | 36.191.54 | 28.330.72 | 75.344.58 |
| 3-CHAIN NLIN | Causal NF | 0.000.00 | 0.030.01 | 0.020.01 | 0.520.06 | 0.560.03 | 1.020.05 |
| | CAREFL | 0.000.00 | 0.050.02 | 0.040.02 | 0.600.22 | 0.840.22 | 1.660.41 |
| | VACA | 12.821.00 | 1.540.03 | 1.320.02 | 39.454.12 | 30.932.30 | 84.369.60 |
| 4-CHAIN LIN | Causal NF | 0.000.00 | 0.070.02 | 0.040.01 | 0.560.08 | 0.620.15 | 1.540.40 |
| | CAREFL | 0.000.00 | 0.160.07 | 0.140.04 | 0.700.28 | 0.990.20 | 2.850.54 |
| | VACA | 13.140.73 | 3.820.01 | 3.720.05 | 61.855.06 | 49.314.11 | 92.067.93 |
| 5-CHAIN LIN | Causal NF | 0.010.00 | 0.120.02 | 0.080.01 | 0.620.19 | 0.690.15 | 1.910.44 |
| | CAREFL | 0.000.00 | 0.470.23 | 0.460.22 | 0.790.41 | 1.190.25 | 4.210.87 |
| | VACA | 17.310.84 | 5.950.05 | 6.060.08 | 103.7510.04 | 80.8111.06 | 124.5220.86 |
| COLLIDER LIN | Causal NF | 0.000.00 | 0.020.01 | 0.010.00 | 0.460.12 | 0.560.11 | 0.950.19 |
| | CAREFL | 0.000.00 | 0.020.01 | 0.010.00 | 0.390.07 | 0.450.05 | 0.740.07 |
| | VACA | 13.450.43 | 0.220.01 | 0.860.02 | 37.223.55 | 28.774.22 | 71.216.73 |



## Fairness Use Case: Partial Graph with SCC Blocks

- We model with partial causal knowledge using SCC grouping of discrete variables.
- We learn ANF within SCCs and preserve inter-block edges; we enable fair interventions on sensitive features.
- We build a foundation for counterfactually fair classification using exogenous variables.

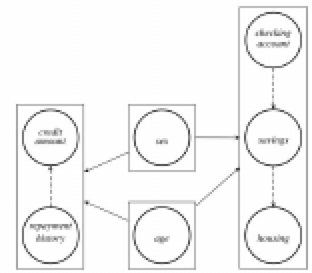


Figure: German Credit: SCC blocks (rectangles), inter-SCC causal edges (solid), and ANF learning order (dashed).



## Fairness Outcomes: Accuracy and Counterfactual Unfairness

- We train classifiers on exogenous $u$ (with sensitive parts removed) and achieve zero counterfactual unfairness.
- We match or exceed the accuracy/F1 of unaware and fair-$x$ baselines.
- We demonstrate practical value of causal NFs for fair decision-making.

Table: Accuracy, F1-score, and counterfactual unfairness of audited classifiers.

| | Logistic classifier | | | | SVM classifier | | | |
|---|---|---|---|---|---|---|---|---|
| | full | unaware | fair x | fair u | full | unaware | fair x | fair u |
| f1 | 72.286.16 | 72.374.90 | 59.668.57 | 73.084.38 | 76.042.86 | 76.805.82 | 68.285.74 | 77.391.52 |
| accuracy | 67.003.83 | 66.752.63 | 54.755.91 | 66.503.70 | 69.503.11 | 71.003.83 | 59.252.99 | 69.751.26 |
| unfairness | 5.842.93 | 2.810.72 | 0.000.00 | 0.000.00 | 6.652.45 | 2.780.40 | 0.000.00 | 0.000.00 |



## Open Questions

- Can we handle hidden confounding with limited interventional data while preserving identifiability?
- How do we relax bijectivity/acyclicity (e.g., cycles, non-bijective mechanisms) without losing causal guarantees?
- Can flow-matching or other training objectives improve data efficiency while maintaining causal consistency?
- What guarantees can we provide under partial graphs and imperfect dequantization in mixed data?
- How do we best deploy causal NFs for fair decision-making and discovery in large domains (e.g., neuroimaging)?

# Questions & Discussion

- Thank you for your attention!
- Questions and feedback are welcome.



# Acknowledgments


- We thank collaborators and the broader causal ML community for foundational ideas.
- Open-source code available to support reproducibility and adoption.




# References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